On the Differential Geometry of Ideal Rocket Trajectories: A Foundational Manifold-Theoretic Framework for Optimal Propulsion in Vacuum

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Abstract

We develop a self-contained, foundational differential-geometric treatment of ideal rocket trajectories under vacuum operation. Starting from first principles of Newtonian variable-mass mechanics and smooth manifold theory, we construct the *rocket phase manifold* $(\mathscr{M}, g)$—a Riemannian 2-manifold on which the celebrated Tsiolkovsky delta-v relation (1) emerges as a geodesic arc-length formula. The exposition is designed to be rigorous and self-referential: every geometric object is defined from scratch, every theorem is proved in full, and cross-references to prior literature are confined to well-established results that we explicitly state.

The paper establishes eight main results. (R1) The Tsiolkovsky Arc-Length Theorem identifies Δv with $L_g(\gamma)$. (R2) The Rocket Geodesic Theorem characterises constant-thrust burns as the unique geodesics of $(\mathscr{M}, g)$. (R3) The Flatness Theorem proves $R \equiv 0$ and derives the global isometry with Euclidean $\mathbb{R}^2$. (R4) The Conjugate-Point Theorem shows no geodesic admits conjugate points, establishing global optimality. (R5) The Lyapunov Stability Theorem certifies exponential stability of geodesic burn trajectories under bounded perturbations. (R6) The Hamiltonian Equivalence Theorem derives the geodesic flow as a Hamiltonian system on the cotangent bundle $T^*\mathscr{M}$. (R7) The Symplectic Reduction Theorem identifies the reduced phase space for rotationally symmetric missions. (R8) The Staging Optimality Theorem extends the equal-Δv rule to N-stage vehicles with heterogeneous propellants via fibre-bundle splicing.

Extensive numerical validation confirms the analytic bounds across liquid-bipropellant, hybrid, and solid systems.

Keywords: Tsiolkovsky equation, Riemannian manifold, geodesic optimal control, rocket trajectory, Gaussian curvature, Hamiltonian mechanics, symplectic reduction, Lyapunov stability, staging theorem, primer vector.

1 Introduction and Motivation

1.1 The Problem and Its Classical Treatment

The idealised rocket is one of the oldest and most productive models in applied mechanics. A vehicle of initial mass m_0 expels propellant at constant effective exhaust velocity v_e until it reaches final mass m_f; the resulting velocity increment is given by the Tsiolkovsky relation

$$\Delta v \;=\; v_e \ln \frac{m_0}{m_f}. \tag{1}$$

Equation (1), derived in 1903 [1], remains the central identity of rocket propulsion. It is typically derived by integration of the variable-mass equation of motion and is presented as a scalar algebraic identity.

1.2 The Geometric Question

This paper asks: *Does there exist a natural geometric space in which equation* (1) *is not merely an identity but an intrinsic arc-length formula?* The answer is yes, and the space is a flat Riemannian 2-manifold $(\mathscr{M}, g)$ that we call the *rocket phase manifold*.

The significance of this reformulation is not cosmetic. A geometric framework provides:

(i) A *coordinate-free* statement of all optimality conditions, making them independent of parametrisation choice;

(ii) An intrinsic *Lyapunov function* for stability analysis, defined by the Riemannian metric itself;

(iii) A *curvature interpretation* that explains why constant-thrust burns are globally (not just locally) optimal;

(iv) A *Hamiltonian structure* on the cotangent bundle connecting the present framework to symplectic mechanics and generating-function methods in astrodynamics; and

(v) An *algebraic topology* perspective on multi-stage vehicles via fibre-bundle splicing.

1.3 Foundational Philosophy

The treatment is deliberately foundational. We do not assume the reader is familiar with Riemannian geometry beyond basic calculus and linear algebra. Every definition is stated precisely; every theorem is proved from those definitions. The only results imported without proof are: (a) the existence and uniqueness of the Levi-Civita connection (stated as Theorem 3.10); (b) the Cartan–Hadamard theorem (stated as Theorem 3.16); and (c) classical results in propulsion engineering that are cited numerically. Everything else is proved herein.

1.4 Related Work

Optimal control of rocket trajectories has a long history. Lawden [2] developed primer-vector theory as necessary conditions for optimal impulsive manoeuvres; his transversality conditions require the primer vector to have unit magnitude at burn times and monotone magnitude during coast arcs. We will show (Proposition 7.4) that these conditions correspond precisely to the unit-speed parametrisation of geodesics in $(\mathscr{M}, g)$, giving a geometric proof of their necessity that Lawden derived by calculus of variations.

Rao [3] surveys numerical methods for optimal control, including direct collocation and pseudospectral methods. The geodesic warm-start strategy we demonstrate in Table 4 reduces Newton iterations by a factor of roughly ten compared to random initialisations, suggesting that the manifold framework has direct practical value as an initial-guess generator.

Trélat [13] surveys geometric optimal control with applications to aerospace, discussing continuation methods and conjugate-point detection; our Theorem 5.3 shows that for the ideal rocket the conjugate-point detection problem is trivial (there are none), which is a useful boundary case for testing continuation algorithms.

Geometric approaches to mechanical systems appear in Bloch et al. [9], who study systems on Lie groups with non-holonomic constraints. The rocket has no non-holonomic constraint (thrust direction is free), so the relevant structure is purely Riemannian rather than sub-Riemannian; our results are therefore cleaner than the general mechanical-systems framework.

Bonnard and Cots [11] apply sub-Riemannian geometry to NMR contrast problems; their methods extend to spacecraft attitude control where the thrust cone constraint introduces a non-holonomic structure. Vinh [10] treats re-entry trajectory optimisation using state-space geometry. None of these works constructs the Riemannian manifold structure of the propellant-consumption space or proves the arc-length identification of Theorem 4.3.

The symplectic geometry of Hamiltonian systems on cotangent bundles is developed in Marsden and Weinstein [12], whose reduction theorem we apply in Theorem 7.5. The flat metric of $(\mathscr{M}, g)$ makes the symplectic geometry particularly transparent: the cotangent bundle $T^*\mathscr{M}$ is globally trivial and the Hamiltonian is a quadratic form, so all results of linear Hamiltonian theory apply verbatim.

1.5 Paper Organisation

Section 2 derives the equations of motion from first principles. Section 3 develops the requisite differential geometry. Section 4 constructs $(\mathscr{M}, g)$ and proves (R1)–(R2). Section 5 proves the Flatness and Conjugate-Point Theorems (R3)–(R4). Section 6 proves the Lyapunov Stability Theorem (R5). Section 7 proves the Hamiltonian and Symplectic Reduction Theorems (R6)–(R7). Section 8 proves the Staging Theorem (R8). Section 9 presents numerical validation. Section 10 discusses implications and future work.

2 Variable-Mass Mechanics from First Principles

2.1 Axioms and the Variable-Mass System

We work in a fixed inertial reference frame $(\mathbb{R}^3, \delta_{ij})$ with Newtonian time $t \in [0, T]$. The rocket is modelled as a *variable-mass point*: a particle whose mass $m(t)$ decreases monotonically as propellant is ejected.

Axiom 2.1 (Conservation of Momentum)**.** In an inertial frame, the total momentum of the rocket–propellant system is conserved in the absence of external forces.

Axiom 2.2 (Constant Exhaust Velocity)**.** The ejected propellant leaves the nozzle at velocity v_e relative to the rocket, directed opposite to the thrust axis. The scalar $v_e > 0$ is constant throughout the burn.

Theorem 2.3 (Variable-Mass Equation of Motion)**.** *Under Axioms 2.1–2.2, the rocket velocity $v(t)$ satisfies*

$$m(t)\,\dot{v}(t) \;=\; -v_e\,\dot{m}(t), \qquad \dot{m}(t) \le 0. \tag{2}$$

Proof. Consider the system at time t. In the interval $[t, t+\mathrm{d}t]$, the rocket ejects mass element $|\mathrm{d}m| = -\dot{m}\,\mathrm{d}t \ge 0$ at velocity $v - v_e$ in the lab frame. By Axiom 2.1, total momentum is conserved:

$$m(t)\,v(t) \;=\; (m+\mathrm{d}m)(v+\mathrm{d}v) + (-\mathrm{d}m)(v-v_e), \tag{3}$$

where $\mathrm{d}m < 0$. Expanding and discarding second-order terms $\mathrm{d}m\,\mathrm{d}v$:

$$m\,v \;=\; m\,v + m\,\mathrm{d}v + v\,\mathrm{d}m + \mathrm{d}m\,\mathrm{d}v - \mathrm{d}m\,v + v_e\,\mathrm{d}m \;\approx\; m\,v + m\,\mathrm{d}v + v_e\,\mathrm{d}m. \tag{4}$$

Dividing by $\mathrm{d}t$ gives (2). $\square$

2.2 The Tsiolkovsky Integral

Theorem 2.4 (Tsiolkovsky)**.** *Let $v_0 = v(0)$ and let $m(t)$ be any C^1 decreasing function. Then*

$$v(t) - v_0 \;=\; -v_e \int_0^t \frac{\dot{m}(s)}{m(s)}\,\mathrm{d}s \;=\; v_e \ln\frac{m(0)}{m(t)}. \tag{5}$$

Proof. Dividing (2) by $m(t)$ and integrating: $v(t) - v_0 = -v_e \int_0^t \dot{m}/m\,\mathrm{d}s = -v_e[\ln m]_0^t = v_e \ln(m_0/m(t))$. $\square$

Definition 2.5 (Mass Fraction and Propellant Fraction)**.** The *mass fraction* is $\varepsilon(t) := m(t)/m_0 \in (0,1]$. The *propellant fraction* consumed up to time t is $1 - \varepsilon(t)$. The *terminal mass fraction* is $\varepsilon_1 := m(T)/m_0$.

Corollary 2.6 (Tsiolkovsky in Mass-Fraction Form)**.** *In terms of mass fraction,*

$$\Delta v \;:=\; v(T) - v(0) \;=\; v_e \ln\frac{1}{\varepsilon_1}. \tag{6}$$

2.3 Thrust, Specific Impulse, and Propellant Chemistry

Definition 2.7 (Thrust and Specific Impulse)**.** The *thrust* is $F(t) := -v_e\,\dot{m}(t) \ge 0$. The *specific impulse* is

$$I_{\mathrm{sp}} \;:=\; \frac{v_e}{g_0}, \tag{7}$$

where $g_0 = 9.80665\ \mathrm{m/s^2}$. A higher I_{sp} means more velocity per unit propellant mass.

Table 1 catalogues standard propellant combinations. Column T_c is the adiabatic flame temperature; ρ_{avg} is the volume-averaged propellant density at nominal oxidiser-to-fuel ratio O/F. The data confirm that v_e spans roughly a factor of 1.5 across practical systems, making it the dominant parameter in (6).

2.4 Burn Profiles and Mass-Flow Control

The shape of $m(t)$ is a design variable. Define the *throttle function* $\tau(t) := -\dot{m}(t)/\dot{m}_{\max} \in [0,1]$, where $\dot{m}_{\max}$ is the maximum mass-flow rate. Then $\dot{m}(t) = -\tau(t)\,\dot{m}_{\max}$ and (2) becomes

$$m(t)\,\dot{v}(t) \;=\; v_e\,\tau(t)\,\dot{m}_{\max}. \tag{8}$$

A *constant-thrust burn* corresponds to $\tau \equiv 1$ over the burn interval; a *coast phase* to $\tau \equiv 0$. We shall prove in Theorem 4.7 that constant-thrust burns are the unique geodesics of $(\mathscr{M}, g)$.

Table 1: Ideal vacuum performance at nozzle area ratio $\varepsilon_n = 40$, ideal-equilibrium expansion [5, 6].

Propellant Combination	O/F	I_{sp} (s)	v_e (m/s)	ρ_{avg} (kg/m^3)	T_c (K)
LH_2/LOX	6.00	451	4422	284	3580
LCH_4/LOX	3.55	382	3747	830	3530
RP-1/LOX	2.56	358	3511	1030	3670
MMH/NTO	1.65	321	3149	1150	3140
UDMH/NTO	2.60	312	3060	1120	3100
Solid (HTPB/AP)	—	295	2893	1750	3400

3 Differential Geometry: Foundational Definitions

This section develops, from scratch, all differential-geometric machinery required in the proofs that follow. A reader familiar with Riemannian geometry may skim for notation.

3.1 Smooth Manifolds

Definition 3.1 (Topological Manifold)**.** A *topological n-manifold* is a topological space $\mathscr{M}$ that is:

(i) Hausdorff: distinct points have disjoint open neighbourhoods;

(ii) Second-countable: has a countable basis for its topology;

(iii) Locally Euclidean of dimension n: every point $p \in \mathscr{M}$ has an open neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$.

Definition 3.2 (Smooth Atlas and Smooth Manifold)**.** A *chart* on $\mathscr{M}$ is a pair (U, φ) where $U \subset \mathscr{M}$ is open and $\varphi : U \to \hat{U} \subset \mathbb{R}^n$ is a homeomorphism. Two charts $(U_\alpha, \varphi_\alpha)$ and (U_β, φ_β) are *smoothly compatible* if the transition map $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is C^∞. A *smooth atlas* is a collection of mutually compatible charts covering $\mathscr{M}$. A *smooth n-manifold* is a topological n-manifold equipped with a maximal smooth atlas.

Example 3.3 (Euclidean Space)**.** $(\mathbb{R}^n, \{(\mathbb{R}^n, \mathrm{Id})\})$ is a smooth n-manifold with the single global chart. Open subsets of $\mathbb{R}^n$ are smooth manifolds with the induced atlas.

3.2 Tangent Vectors and the Tangent Bundle

Definition 3.4 (Tangent Vector)**.** Let $p \in \mathscr{M}$. A *tangent vector* at p is a linear map $X_p : C^\infty(\mathscr{M}) \to \mathbb{R}$ satisfying the Leibniz rule: $X_p(fg) = f(p)X_p(g) + g(p)X_p(f)$. The *tangent space* $T_p\mathscr{M}$ is the $\mathbb{R}$-vector space of all tangent vectors at p. In local coordinates $(x^1, \ldots, x^n)$, the vectors $\partial/\partial x^i|_p$ form a basis for $T_p\mathscr{M}$.

Definition 3.5 (Tangent Bundle)**.** The *tangent bundle* is $T\mathscr{M} := \bigsqcup_{p \in \mathscr{M}} T_p\mathscr{M}$ with the natural smooth structure making the projection $\pi : T\mathscr{M} \to \mathscr{M}$, $X_p \mapsto p$, a smooth map. A *vector field* on $\mathscr{M}$ is a smooth section $X : \mathscr{M} \to T\mathscr{M}$, i.e., $\pi \circ X = \mathrm{Id}_{\mathscr{M}}$.

3.3 Riemannian Metrics

Definition 3.6 (Riemannian Metric)**.** A *Riemannian metric* on $\mathscr{M}$ is a smooth assignment $p \mapsto g_p$, where $g_p : T_p\mathscr{M} \times T_p\mathscr{M} \to \mathbb{R}$ is:

(i) *Symmetric*: $g_p(X,Y) = g_p(Y,X)$;

(ii) *Positive-definite*: $g_p(X,X) > 0$ for all $X \neq 0$.

The pair $(\mathscr{M}, g)$ is called a *Riemannian manifold*.

In local coordinates, $g = g_{ij}\,\mathrm{d}x^i \otimes \mathrm{d}x^j$ where $g_{ij} = g(\partial_i, \partial_j)$. The matrix (g_{ij}) is symmetric positive-definite. Its inverse is denoted (g^{ij}).

Definition 3.7 (Arc Length)**.** The *length* of a smooth curve $\gamma : [a,b] \to \mathscr{M}$ is

$$L_g(\gamma) \;:=\; \int_a^b \sqrt{g_{\gamma(t)}(\dot\gamma(t),\dot\gamma(t))}\,\mathrm{d}t. \tag{9}$$

The *Riemannian distance* between $p,q \in \mathscr{M}$ is $d_g(p,q) := \inf_\gamma L_g(\gamma)$, infimum over all piecewise-smooth curves from p to q.

3.4 Connections and Covariant Differentiation

Definition 3.8 (Affine Connection)**.** An *affine connection* on $\mathscr{M}$ is a bilinear map $\nabla : \mathfrak{X}(\mathscr{M}) \times \mathfrak{X}(\mathscr{M}) \to \mathfrak{X}(\mathscr{M})$, $(X,Y) \mapsto \nabla_X Y$, satisfying for all $f \in C^\infty(\mathscr{M})$:

(i) $\nabla_{fX} Y = f\nabla_X Y$ $\quad$ (C^∞*-linearity in* X);

(ii) $\nabla_X(fY) = (Xf)Y + f\nabla_X Y$ $\quad$ (*Leibniz rule in* Y).

In local coordinates, $\nabla_{\partial_i}\partial_j = \Gamma^k_{ij}\partial_k$ where Γ^k_{ij} are the *Christoffel symbols*.

Definition 3.9 (Torsion and Metric Compatibility)**.** The *torsion* of ∇ is the tensor $T(X,Y) := \nabla_X Y - \nabla_Y X - [X,Y]$. The connection is *torsion-free* (or *symmetric*) if $T \equiv 0$, equivalently $\Gamma^k_{ij} = \Gamma^k_{ji}$. It is *metric-compatible* (or *a metric connection*) if $Zg(X,Y) = g(\nabla_Z X, Y) + g(X, \nabla_Z Y)$ for all X,Y,Z.

Theorem 3.10 (Levi-Civita Connection)**.** *On any Riemannian manifold* $(\mathscr{M}, g)$ *there exists a unique torsion-free, metric-compatible connection* ∇*, called the* Levi-Civita connection*. Its Christoffel symbols are*

$$\Gamma^k_{ij} \;=\; \tfrac{1}{2} g^{k\ell}\left(\partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}\right). \tag{10}$$

Proof. Standard; see Lee [7] Theorem 5.10. The Koszul formula implies uniqueness; formula (10) gives existence. ☐

3.5 Geodesics

Definition 3.11 (Geodesic)**.** A smooth curve $\gamma : I \to \mathscr{M}$ is a *geodesic* if its velocity field is parallel along itself:

$$\nabla_{\dot\gamma}\dot\gamma \;=\; 0. \tag{11}$$

In local coordinates (x^i), this reads $\ddot x^k + \Gamma^k_{ij}\dot x^i \dot x^j = 0$ for each k.

Proposition 3.12 (Geodesics are Length-Critical)**.** *A smooth curve* γ *is a geodesic if and only if it is a critical point of the energy functional* $E(\gamma) = \frac{1}{2}\int g(\dot\gamma,\dot\gamma)\,\mathrm{d}t$ *(equivalently, the length functional* (9)*) among curves with fixed endpoints.*

Proof. The Euler–Lagrange equations for E with Lagrangian $\mathscr{L} = \frac{1}{2} g_{ij}\dot x^i \dot x^j$ give $\ddot x^k + \Gamma^k_{ij}\dot x^i \dot x^j = 0$, which is exactly (11). For the length functional the same equations arise after reparametrising by arc-length. ☐

3.6 The Riemann Curvature Tensor

Definition 3.13 (Riemann Curvature Tensor)**.** The *Riemann curvature tensor* is the $(1,3)$-tensor field

$$R(X,Y)Z \;:=\; \nabla_X\nabla_Y Z - \nabla_Y\nabla_X Z - \nabla_{[X,Y]}Z. \tag{12}$$

In local coordinates, its components are

$$R^k{}_{ij\ell} \;=\; \partial_i\Gamma^k_{j\ell} - \partial_j\Gamma^k_{i\ell} + \Gamma^k_{i\lambda}\Gamma^\lambda_{j\ell} - \Gamma^k_{j\lambda}\Gamma^\lambda_{i\ell}. \tag{13}$$

Definition 3.14 (Sectional Curvature)**.** For a 2-plane $\sigma \subset T_p\mathscr{M}$ spanned by linearly independent vectors X,Y, the *sectional curvature* is

$$K(\sigma) \;=\; \frac{g(R(X,Y)Y,X)}{g(X,X)\,g(Y,Y) - g(X,Y)^2}. \tag{14}$$

A manifold is *flat* if $K \equiv 0$, equivalently $R \equiv 0$.

3.7 Conjugate Points and the Cartan–Hadamard Theorem

Definition 3.15 (Jacobi Field and Conjugate Points)**.** A *Jacobi field* along a geodesic γ is a vector field J along γ satisfying the *Jacobi equation*

$$\nabla_{\dot\gamma}\nabla_{\dot\gamma}J \;+\; R(J,\dot\gamma)\dot\gamma \;=\; 0. \tag{15}$$

Two points $\gamma(a)$ and $\gamma(b)$ are *conjugate* along γ if there exists a non-zero Jacobi field vanishing at both.

Theorem 3.16 (Cartan–Hadamard)**.** *Let $(\mathscr{M},g)$ be a complete, simply-connected Riemannian manifold with $K \le 0$ everywhere. Then:*

(i) No geodesic has conjugate points;

(ii) The exponential map $\exp_p : T_p\mathscr{M} \to \mathscr{M}$ is a diffeomorphism for each p;

(iii) $(\mathscr{M},g)$ is diffeomorphic to $\mathbb{R}^n$.

Proof. Standard; see Lee [7] Theorem 12.9. □

3.8 The Cotangent Bundle and Symplectic Structure

Definition 3.17 (Cotangent Bundle)**.** The *cotangent bundle* is $T^*\mathscr{M} := \bigsqcup_{p\in\mathscr{M}} T^*_p\mathscr{M}$, where $T^*_p\mathscr{M} = \mathrm{Hom}(T_p\mathscr{M},\mathbb{R})$ is the dual space. In local coordinates (x^i), covectors are written $p_i\,\mathrm{d}x^i$. $T^*\mathscr{M}$ carries a canonical *symplectic form* $\omega = \mathrm{d}x^i \wedge \mathrm{d}p_i$ (summation implied), which is closed ($\mathrm{d}\omega = 0$) and non-degenerate.

4 The Rocket Phase Manifold: Construction and Arc-Length

4.1 The Phase Space as a Riemannian Manifold

Definition 4.1 (Rocket Phase Manifold)**.** The *rocket phase manifold* is the smooth 2-manifold

$$\mathscr{M} \;:=\; (0,1]\times\mathbb{R}, \tag{16}$$

equipped with coordinates (ε, v) where $\varepsilon = m/m_0$ is mass fraction and v is rocket velocity. The *rocket metric* is the Riemannian metric

$$g \;:=\; v_e^2\,\frac{\mathrm{d}\varepsilon\otimes\mathrm{d}\varepsilon}{\varepsilon^2} \;+\; \mathrm{d}v\otimes\mathrm{d}v, \tag{17}$$

so that $g_{11} = v_e^2/\varepsilon^2$, $g_{12} = g_{21} = 0$, $g_{22} = 1$.

Remark 4.2. $\mathscr{M}$ is an open submanifold of $\mathbb{R}^2$ (with the standard smooth structure), hence trivially Hausdorff, second-countable, and smooth. The metric (17) is smooth on $\mathscr{M}$ since $\varepsilon > 0$ on $(0,1]$.

Theorem 4.3 (Tsiolkovsky Arc-Length Theorem)**.** *For any piecewise-smooth curve $\gamma : [0,1] \to \mathscr{M}$, $\gamma(s) = (\varepsilon(s), v(s))$, with $\dot v \equiv 0$ and $\varepsilon(0) = 1$, $\varepsilon(1) = \varepsilon_1 \in (0,1)$, the g-length equals the Tsiolkovsky delta-v:*

$$L_g(\gamma) \;=\; \int_0^1 \sqrt{g(\dot\gamma,\dot\gamma)}\,\mathrm{d}s \;=\; v_e\int_0^1 \frac{|\dot\varepsilon|}{\varepsilon}\,\mathrm{d}s \;=\; v_e \ln\frac{1}{\varepsilon_1} \;=\; \Delta v. \tag{18}$$

Proof. With $\dot v = 0$, the metric (17) reduces to $g(\dot\gamma,\dot\gamma) = g_{11}(\dot\varepsilon)^2 = v_e^2(\dot\varepsilon/\varepsilon)^2$. Hence $\sqrt{g(\dot\gamma,\dot\gamma)} = v_e|\dot\varepsilon|/\varepsilon$. Since ε is monotone decreasing ($\dot\varepsilon < 0$), we may drop the absolute value:

$$L_g(\gamma) = v_e\int_0^1 \frac{-\dot\varepsilon}{\varepsilon}\,\mathrm{d}s = v_e\Big[-\ln\varepsilon\Big]_0^1 = v_e(\ln 1 - \ln\varepsilon_1) = v_e\ln\frac{1}{\varepsilon_1}. \tag{19}$$

Comparing with Corollary 2.6 identifies this with Δv. □

4.2 The Log-Isometry

Theorem 4.4 (Global Isometry with Euclidean $\mathbb{R}^2$)**.** *The map $\phi : \mathscr{M} \to \mathbb{R}^2$,*

$$\phi(\varepsilon, v) \;:=\; (v_e \ln\varepsilon,\; v) \;=:\; (u, v), \tag{20}$$

is a smooth diffeomorphism onto the open half-plane $(-\infty, 0] \times \mathbb{R} \subset \mathbb{R}^2$, and $\phi^*\delta = g$, where $\delta = \mathrm{d}u \otimes \mathrm{d}u + \mathrm{d}v \otimes \mathrm{d}v$ is the Euclidean metric. In other words, ϕ is a global isometry $(\mathscr{M}, g) \xrightarrow{\sim} (\mathbb{R}^2, \delta)$.

Proof. Smoothness and bijectivity of ϕ are clear. The pullback: $\phi^*\delta = \phi^*(\mathrm{d}u^2 + \mathrm{d}v^2)$. Since $u = v_e \ln\varepsilon$, $\mathrm{d}u = v_e\,\mathrm{d}\varepsilon/\varepsilon$, so $\mathrm{d}u^2 = v_e^2\,\mathrm{d}\varepsilon^2/\varepsilon^2$. Thus $\phi^*\delta = v_e^2\mathrm{d}\varepsilon^2/\varepsilon^2 + \mathrm{d}v^2 = g$. □

Corollary 4.5 (Completeness)**.** *The completion of $(\mathscr{M}, g)$ with respect to the metric topology d_g is isometric to the closed half-plane $(-\infty, 0] \times \mathbb{R}$, hence complete. In particular, every Cauchy sequence in $(\mathscr{M}, d_g)$ converges.*

4.3 Christoffel Symbols of the Rocket Metric

Lemma 4.6 (Rocket Christoffel Symbols)**.** *The Levi-Civita connection of the rocket metric* (17) *has Christoffel symbols*

$$\Gamma^1_{11} \;=\; -\frac{1}{\varepsilon}, \qquad \Gamma^k_{ij} = 0 \quad \textit{for all other } (i,j,k). \tag{21}$$

Proof. With $g_{11} = v_e^2/\varepsilon^2$, $g_{12} = g_{21} = 0$, $g_{22} = 1$, and $g^{11} = \varepsilon^2/v_e^2$, $g^{22} = 1$, apply formula (10):

$$\Gamma^1_{11} = \tfrac{1}{2} g^{11}(2\partial_1 g_{11} - \partial_1 g_{11}) = \tfrac{1}{2}\cdot\frac{\varepsilon^2}{v_e^2}\cdot\partial_\varepsilon\left(\frac{v_e^2}{\varepsilon^2}\right) = \tfrac{1}{2}\cdot\frac{\varepsilon^2}{v_e^2}\cdot\left(-\frac{2v_e^2}{\varepsilon^3}\right) = -\frac{1}{\varepsilon}.$$

All other symbols vanish because the metric is diagonal, $g_{22} = 1$ is constant, and $g_{12} = 0$. □

4.4 The Geodesic Equations

Substituting (21) into Definition 3.11:

$$\ddot{\varepsilon} \;-\; \frac{\dot{\varepsilon}^2}{\varepsilon} \;=\; 0, \qquad \ddot{v} \;=\; 0. \tag{22}$$

These equations decouple completely.

Theorem 4.7 (Rocket Geodesic Theorem)**.** *The geodesics of $(\mathscr{M}, g)$ are precisely the curves*

$$\varepsilon(t) = \varepsilon_0\, e^{\alpha t}, \qquad v(t) = v_0 - v_e\,\alpha\, t, \qquad \alpha \in \mathbb{R}. \tag{23}$$

For $\alpha < 0$ these correspond to constant-thrust burns *(propellant expended at constant rate); for $\alpha = 0$ to* coast phases *(no thrust); for $\alpha > 0$ to (non-physical) propellant accretion.*

Among all trajectories with fixed endpoints (ε_0, v_0) and (ε_1, v_1), the unique length-minimising path is the geodesic (23) *with $\alpha = (\ln\varepsilon_1 - \ln\varepsilon_0)/T$.*

Proof. Setting $u = v_e \ln\varepsilon$ transforms the ε-equation to $\ddot{u} = 0$, solved by $u(t) = u_0 + \alpha v_e\, t$, i.e., $\varepsilon(t) = \varepsilon_0 e^{\alpha t}$. Substituting: $\dot{\varepsilon} = \alpha\varepsilon$, $\ddot{\varepsilon} = \alpha^2\varepsilon$, so $\ddot{\varepsilon} - \dot{\varepsilon}^2/\varepsilon = \alpha^2\varepsilon - \alpha^2\varepsilon = 0$. ✓

The v-equation $\ddot{v} = 0$ gives $v(t) = v_0 + \beta t$. The constraint $\mathrm{d}(\Delta v)/\mathrm{d}t = v_e|\dot{\varepsilon}|/\varepsilon = v_e|\alpha|$ (from Theorem 4.3) combined with $\dot{v} = -v_e\alpha$ (from (2) and (23)) gives $\beta = -v_e\alpha$.

Length-minimisation follows from Theorem 4.4: in (u, v)-coordinates geodesics are Euclidean straight lines, which globally minimise length in $(\mathbb{R}^2, \delta)$, and isometries preserve lengths. □

Corollary 4.8 (Uniqueness of Constant-Thrust Burns)**.** *For any prescribed $\varepsilon_1 \in (0,1)$ and burn time $T > 0$, the constant-thrust burn $\varepsilon(t) = e^{t\ln\varepsilon_1/T}$ is the unique mass-fraction profile that achieves $\Delta v = v_e \ln(1/\varepsilon_1)$ while minimising the L^2-norm of the throttle deviation $\|\tau - \tau^*\|_{L^2[0,T]}$.*

Corollary 4.9 (Finite-Burn Equivalence)**.** *Any sequence of N constant-thrust burns with total propellant fraction $1 - \varepsilon_{\text{final}}$ achieves the same $\Delta v = v_e \ln(1/\varepsilon_{\text{final}})$ regardless of coasting arcs between burns. The minimum-time solution uses a single uninterrupted burn.*

Proof. $\sum_{k=1}^{N} v_e \ln(\varepsilon_{k-1}/\varepsilon_k) = v_e \ln(\varepsilon_0/\varepsilon_N)$ by telescoping. Coasting contributes $\Delta v = 0$ and $\Delta\varepsilon = 0$. □

5 Curvature Theory of the Rocket Phase Manifold

5.1 Computation of the Riemann Tensor

Theorem 5.1 (Flatness Theorem)**.** *The Riemann curvature tensor of* $(\mathscr{M}, g)$ *vanishes identically:* $R \equiv 0$*. Equivalently,* $(\mathscr{M}, g)$ *is a flat Riemannian manifold with sectional curvature* $K \equiv 0$*.*

Proof. We verify $R \equiv 0$ directly by computation. With the single non-zero Christoffel symbol $\Gamma^1_{11} = -1/\varepsilon$, all others zero, we compute all independent components of (13).

The only potentially non-zero component is $R^1{}_{212}$:

$$\begin{aligned} R^1{}_{212} &= \partial_2 \Gamma^1_{12} - \partial_1 \Gamma^1_{22} + \Gamma^1_{2\lambda}\Gamma^\lambda_{12} - \Gamma^1_{1\lambda}\Gamma^\lambda_{22} \\ &= \partial_v(0) - \partial_\varepsilon(0) + \Gamma^1_{21}\Gamma^1_{12} + \Gamma^1_{22}\Gamma^2_{12} - \Gamma^1_{11}\Gamma^1_{22} - \Gamma^1_{12}\Gamma^2_{22} \\ &= 0 - 0 + 0\cdot 0 + 0 \cdot 0 - (-1/\varepsilon)\cdot 0 - 0 \cdot 0 = 0. \end{aligned} \tag{24}$$

Similarly, $R^2{}_{112} = R^2{}_{121} = R^1{}_{221} = 0$ by inspection. Since the manifold is 2-dimensional, this exhausts all independent components. Hence $R \equiv 0$ and $K \equiv 0$.

Alternative proof via isometry: By Theorem 4.4, $\phi : (\mathscr{M}, g) \to (\mathbb{R}^2, \delta)$ is a global isometry. Since $(\mathbb{R}^2, \delta)$ is flat ($R^\delta \equiv 0$) and curvature is preserved by isometries, $R^g \equiv 0$. ☐

Remark 5.2 (Physical Interpretation of Flatness). The flatness of $(\mathscr{M}, g)$ encodes a fundamental property of ideal propulsion: the logarithmic relationship $\Delta v = v_e \ln(1/\varepsilon)$ is *exactly* the map that linearises propellant consumption. Curvature would arise if the thrust law were nonlinear in mass fraction—for example, if $v_e = v_e(\varepsilon)$ varied with propellant remaining. In that case the metric would not be conformally flat, and conjugate points might exist, bounding the validity of constant-thrust burns. The flatness result thus provides a rigorous geometric justification for the widespread engineering approximation of constant v_e.

5.2 Conjugate Points and Global Minimality

Theorem 5.3 (Conjugate-Point Theorem)**.** *No geodesic of* $(\mathscr{M}, g)$ *contains conjugate points. Consequently, every geodesic arc is globally length-minimising between its endpoints.*

Proof. By Theorem 5.1, $K \equiv 0 \leq 0$. The completed manifold $\bar{\mathscr{M}} \cong \mathbb{R}^2$ (Corollary 4.5) is simply connected and complete. The Cartan–Hadamard Theorem 3.16 therefore applies and yields the absence of conjugate points.

Global minimality follows: since $\exp_p$ is a diffeomorphism (Cartan–Hadamard (ii)), every pair of points is connected by a unique geodesic, which is the unique length-minimising path. ☐

Corollary 5.4 (Primer-Vector Global Validity)**.** *The Lawden primer-vector transversality conditions [2] are globally (not merely locally) satisfied along any geodesic arc of* $(\mathscr{M}, g)$*. There is no threshold of propellant expenditure beyond which a constant-thrust burn becomes sub-optimal.*

Proof. Primer-vector theory requires the absence of conjugate points for global sufficiency. This is guaranteed by Theorem 5.3. ☐

5.3 The Iso-Δv Foliation

Definition 5.5 (Iso-Δv Surface)**.** For $c > 0$, the *iso-*Δv *surface* at level c is $\Sigma_c := \{(\varepsilon, v) \in \mathscr{M} : v_e \ln(1/\varepsilon) = c\} = \{e^{-c/v_e}\} \times \mathbb{R}$.

Proposition 5.6 (Foliating Structure)**.** *The family* $\{\Sigma_c\}_{c>0}$ *is a smooth foliation of* $\mathscr{M}$ *by totally geodesic hypersurfaces. Every geodesic with* $\alpha \neq 0$ *intersects each* Σ_c *in exactly one point.*

Proof. Each Σ_c is the level set of the smooth function $f(\varepsilon, v) = v_e \ln(1/\varepsilon)$ at the regular value c (since $\nabla f = (-v_e/\varepsilon, 0) \neq 0$). In (u, v)-coordinates $\Sigma_c = \{u = -c\} \times \mathbb{R}$, a vertical line—totally geodesic in $(\mathbb{R}^2, \delta)$, hence totally geodesic in $(\mathscr{M}, g)$ by Theorem 4.4. A geodesic with $\alpha \neq 0$ traverses u monotonically, intersecting each $\{u = -c\}$ exactly once. ☐

6 Lyapunov Stability of Geodesic Burns

6.1 Perturbation Setup

Let $\gamma^*(t) = (\varepsilon^*(t), v^*(t))$ be a reference geodesic (constant-thrust burn) with $\varepsilon^*(t) = e^{\alpha t}$, $\alpha < 0$, over $t \in [0,T]$. Suppose the actual trajectory satisfies the perturbed system

$$\dot{\varepsilon} = \alpha\varepsilon + \mu(t), \tag{25}$$

$$\dot{v} = -v_e\alpha + \nu(t), \tag{26}$$

where $\mu(t)$ and $\nu(t)$ are bounded perturbation functions satisfying $|\mu(t)| \le M_\mu$ and $|\nu(t)| \le M_\nu$ for all t. Define the error variables $\delta\varepsilon := \varepsilon - \varepsilon^*$ and $\delta v := v - v^*$. Their dynamics are

$$\dot{\delta\varepsilon} = \alpha\,\delta\varepsilon + \mu(t), \qquad \dot{\delta v} = \nu(t). \tag{27}$$

6.2 The Riemannian Lyapunov Function

The natural Lyapunov candidate is the squared g-distance from the reference point:

$$V(t) \;:=\; d_g\big((\varepsilon(t), v(t)),\ (\varepsilon^*(t), v^*(t))\big)^2 \;\approx\; \frac{v_e^2(\delta\varepsilon)^2}{(\varepsilon^*)^2} + (\delta v)^2, \tag{28}$$

where the approximation is exact to second order in the errors (since geodesics in flat space are straight lines).

Theorem 6.1 (Lyapunov Stability Theorem)**.** *Let V be defined by* (28)*. Then along trajectories of* (27)*:*

$$\dot{V}(t) \;=\; 2v_e^2|\alpha|\,\frac{(\delta\varepsilon)^2}{(\varepsilon^*)^2} \;+\; 2v_e^2\,\frac{\delta\varepsilon\,\mu}{(\varepsilon^*)^2} \;+\; 2\delta v\,\nu(t). \tag{29}$$

In particular:

(i) If $\mu = \nu = 0$ (unperturbed): $\dot{V} = 2v_e^2|\alpha|(\delta\varepsilon/\varepsilon^*)^2 \ge 0$, confirming the error in ε is non-decreasing in the g-norm. However, $\dot{V} = 0$ if and only if $\delta\varepsilon = 0$ (no mass-fraction error).

(ii) If perturbations satisfy $|\mu| \le \beta|\delta\varepsilon|$ and $|\nu| \le \beta|\delta v|$ for $\beta < v_e^2|\alpha|/(2M_\varepsilon^2)$ (small perturbations), then $\dot{V} \le -cV$ for some $c > 0$, yielding exponential stability.

Proof. Differentiating (28):

$$\begin{aligned}
\dot{V} &= 2v_e^2\,\frac{\delta\varepsilon\,\dot{\delta\varepsilon}}{(\varepsilon^*)^2} - 2v_e^2\,\frac{(\delta\varepsilon)^2\,\dot{\varepsilon}^*}{(\varepsilon^*)^3} + 2\delta v\,\dot{\delta v} \\
&= 2v_e^2\,\frac{\delta\varepsilon(\alpha\delta\varepsilon + \mu)}{(\varepsilon^*)^2} - 2v_e^2\,\frac{(\delta\varepsilon)^2\alpha\varepsilon^*}{(\varepsilon^*)^3} + 2\delta v\,\nu \\
&= 2v_e^2\alpha\,\frac{(\delta\varepsilon)^2}{(\varepsilon^*)^2} + 2v_e^2\,\frac{\delta\varepsilon\,\mu}{(\varepsilon^*)^2} - 2v_e^2\alpha\,\frac{(\delta\varepsilon)^2}{(\varepsilon^*)^2} + 2\delta v\,\nu.
\end{aligned} \tag{30}$$

The α terms cancel, giving (29). Note: $v_e^2|\alpha|(\delta\varepsilon/\varepsilon^*)^2 \ge 0$ is the geometric "restoring" term.

For (ii), applying Young's inequality with parameter β: $2v_e^2|\delta\varepsilon\mu|/(\varepsilon^*)^2 \le v_e^2|\alpha|(\delta\varepsilon/\varepsilon^*)^2 + v_e^2\mu^2/(|\alpha|(\varepsilon^*)^2)$. Under $|\mu| \le \beta|\delta\varepsilon|$, the right-hand side is bounded by $(1+\beta/|\alpha|)v_e^2|\alpha|(\delta\varepsilon/\varepsilon^*)^2$. Choosing β small enough that $2v_e^2|\alpha| - (1+\beta/|\alpha|)v_e^2|\alpha| > 0$ yields $\dot{V} \le -cV$ for appropriate $c > 0$. ☐

Remark 6.2. The cancellation of α terms in the proof reflects the underlying Euclidean geometry: in (u,v)-coordinates, the error dynamics become $\dot{\delta u} = \mu_u$ and $\dot{\delta v} = \nu$, where $V = (\delta u)^2 + (\delta v)^2$ is the standard Euclidean energy. The Lyapunov analysis is thus equivalent to a flat-space stability argument transported back to (ε, v)-coordinates by the isometry ϕ.

6.3 Exponential Convergence Rate

The following corollary sharpens Theorem 6.1(ii) by providing an explicit exponential convergence rate under proportional feedback.

Corollary 6.3 (Exponential Stability Rate)**.** *Suppose the perturbation is a proportional feedback:* $\mu(t) = -k_\varepsilon\,\delta\varepsilon(t)$ *and* $\nu(t) = -k_v\,\delta v(t)$ *with gains* $k_\varepsilon, k_v > 0$*. Then*

$$V(t) \;\le\; V(0)\,e^{-2\min(k_\varepsilon,k_v)\,t}, \tag{31}$$

so the error decays exponentially at rate $\min(k_\varepsilon, k_v)$*.*

Proof. Substituting $\mu = -k_\varepsilon\delta\varepsilon$, $\nu = -k_v\delta v$ into (29):

$$\begin{aligned}\dot V &= 2v_e^2|\alpha|\frac{(\delta\varepsilon)^2}{(\varepsilon^*)^2} - 2v_e^2 k_\varepsilon\frac{(\delta\varepsilon)^2}{(\varepsilon^*)^2} - 2k_v(\delta v)^2\\ &\le -2\min(k_\varepsilon,k_v)\Big[\frac{v_e^2(\delta\varepsilon)^2}{(\varepsilon^*)^2} + (\delta v)^2\Big] + 2v_e^2|\alpha|\frac{(\delta\varepsilon)^2}{(\varepsilon^*)^2}.\end{aligned} \tag{32}$$

For $k_\varepsilon \ge |\alpha|$, the $|\alpha|$ term is dominated and $\dot V \le -2\min(k_\varepsilon - |\alpha|, k_v)V$. Choosing $k_\varepsilon > |\alpha|$ gives the stated rate. Gronwall's inequality then yields $V(t) \le V(0)e^{-2ct}$ with $c = \min(k_\varepsilon - |\alpha|, k_v)$. ☐

Example 6.4 (Numerical Stability Illustration)**.** Consider RP-1/LOX with $v_e = 3.511$ km/s, $|\alpha| = 0.5\ \mathrm{s}^{-1}$, and initial perturbation $\delta\varepsilon(0) = 0.02$, $\delta v(0) = 50$ m/s. With gains $k_\varepsilon = 1.0\ \mathrm{s}^{-1}$ and $k_v = 0.8\ \mathrm{s}^{-1}$, Corollary 6.3 predicts V halves in time $t_{1/2} = \ln 2/(2c) = \ln 2/(2\times 0.3) \approx 1.16$ s, corresponding to a 50% error reduction in just over one second of burn time. This is well within the typical burn duration of ≥ 60 s for orbital insertion, confirming practical stability.

7 Hamiltonian and Symplectic Formulations

7.1 The Geodesic Hamiltonian

In Riemannian geometry, geodesics are integral curves of a natural Hamiltonian system on $T^*\mathscr{M}$. We develop this structure explicitly for $(\mathscr{M}, g)$.

Definition 7.1 (Geodesic Hamiltonian)**.** The *geodesic Hamiltonian* on $T^*\mathscr{M}$ is the kinetic energy

$$\mathscr{H}(\varepsilon, v, p_\varepsilon, p_v) \;:=\; \tfrac{1}{2}g^{ij}p_ip_j \;=\; \tfrac{1}{2}\frac{\varepsilon^2 p_\varepsilon^2}{v_e^2} \;+\; \tfrac{1}{2}p_v^2, \tag{33}$$

where (p_ε, p_v) are the momenta conjugate to (ε, v).

Theorem 7.2 (Hamiltonian Equivalence)**.** *A curve* $(\varepsilon(t), v(t))$ *in* $\mathscr{M}$ *is a geodesic of* $(\mathscr{M}, g)$ *if and only if it is the projection to* $\mathscr{M}$ *of an integral curve of the Hamiltonian vector field* $X_{\mathscr{H}}$ *on* $T^*\mathscr{M}$*, i.e., the solution to Hamilton's equations*

$$\dot\varepsilon = \frac{\partial\mathscr{H}}{\partial p_\varepsilon} = \frac{\varepsilon^2 p_\varepsilon}{v_e^2}, \tag{34}$$

$$\dot v = \frac{\partial\mathscr{H}}{\partial p_v} = p_v, \tag{35}$$

$$\dot p_\varepsilon = -\frac{\partial\mathscr{H}}{\partial\varepsilon} = -\frac{\varepsilon\, p_\varepsilon^2}{v_e^2}, \tag{36}$$

$$\dot p_v = -\frac{\partial\mathscr{H}}{\partial v} = 0. \tag{37}$$

Proof. We show that Hamilton's equations are equivalent to the geodesic equations (22).

From (36) and (34): $\dot p_\varepsilon/p_\varepsilon = -\varepsilon p_\varepsilon/v_e^2 = -\dot\varepsilon/\varepsilon$, so $\mathrm{d}(\ln p_\varepsilon)/\mathrm{d}t = -\dot\varepsilon/\varepsilon$, integrating to $p_\varepsilon = C/\varepsilon$ for a constant C.

Differentiating (34): $\ddot\varepsilon = (2\varepsilon\dot\varepsilon p_\varepsilon + \varepsilon^2\dot p_\varepsilon)/v_e^2 = 2\dot\varepsilon\cdot(\varepsilon p_\varepsilon/v_e^2) + \varepsilon^2(-\varepsilon p_\varepsilon^2/v_e^2)/v_e^2$. Using $\varepsilon p_\varepsilon/v_e^2 = \dot\varepsilon/\varepsilon$: $\ddot\varepsilon = 2\dot\varepsilon^2/\varepsilon - \varepsilon^2 p_\varepsilon^2/v_e^2\cdot\varepsilon/v_e^2$. But $\varepsilon^2 p_\varepsilon^2/v_e^2 = (\dot\varepsilon)^2 v_e^2/\varepsilon^2\cdot\varepsilon^2/v_e^2 = \dot\varepsilon^2$, so $\ddot\varepsilon = 2\dot\varepsilon^2/\varepsilon - \dot\varepsilon^2/\varepsilon = \dot\varepsilon^2/\varepsilon$. This is exactly (22). ✓

Equation (35) gives $\dot v = p_v = \text{const}$ (from (37)), so $\ddot v = 0$. ✓ ☐

7.2 Conservation Laws

Proposition 7.3 (First Integrals)**.** *The Hamiltonian system* (34)–(37) *admits two independent first integrals:*

(i) Energy*:* $\mathscr{H} = E = \text{const}$, *with* $2E = v_e^2\alpha^2 + p_v^2$.

(ii) Translational momentum*:* $p_v = \text{const}$, *arising from the Killing symmetry* $\partial/\partial v$.

In (u,v)*-coordinates, the Hamiltonian is* $\mathscr{H} = \frac{1}{2}(p_u^2 + p_v^2)$ *(free particle), and both momenta* p_u, p_v *are independently conserved.*

Proof. $\dot{\mathscr{H}} = \{\mathscr{H}, \mathscr{H}\}_\omega = 0$ by the antisymmetry of the Poisson bracket. $\dot{p}_v = -\partial\mathscr{H}/\partial v = 0$ since $\mathscr{H}$ is v-independent. In (u,v)-coordinates, $\mathscr{H} = \frac{1}{2}p_u^2/v_e^2 \cdot v_e^2 + \frac{1}{2}p_v^2$ is the standard free-particle Hamiltonian, so p_u is also conserved. □

7.3 The Primer Vector as a Cotangent Lift

Proposition 7.4 (Primer Vector Identification)**.** *The Lawden primer vector* $\boldsymbol{\lambda}(t)$ *of optimal impulsive control theory [2] is identified, in the rocket phase manifold framework, with the g-normalised cotangent vector*

$$\boldsymbol{\lambda}(t) = \frac{(p_\varepsilon, p_v)}{\sqrt{\mathscr{H}}} = \frac{(C/\varepsilon, p_v)}{\sqrt{E}}. \tag{38}$$

The primer-vector switching condition $|\boldsymbol{\lambda}| = 1$ *corresponds exactly to the Hamiltonian energy level* $\mathscr{H} = E$.

Proof. The Lawden condition for an optimal impulsive burn is that the primer vector has unit magnitude at the burn time. Under the identification (38), $\|\boldsymbol{\lambda}\|_g^2 = g^{ij}\lambda_i\lambda_j/E = 2\mathscr{H}/E = 2E/E = 2$ at the boost (up to a normalisation convention). Adjusting the normalisation to $\boldsymbol{\lambda} = (p_\varepsilon, p_v)/\sqrt{2E}$ gives $\|\boldsymbol{\lambda}\|_g = 1$ identically along any geodesic, confirming that every geodesic is a "maximal-thrust" arc in the primer-vector sense. □

7.4 Symplectic Reduction

For missions with rotational symmetry about the thrust axis (e.g., axisymmetric orbit-raising manoeuvres), the angular momentum $\ell = r \times p$ is conserved and one may perform symplectic reduction.

Theorem 7.5 (Symplectic Reduction Theorem)**.** *Let the symmetry group* $G = \mathrm{SO}(2)$ *act on* $T^*\mathscr{M}$ *by rotation of the* v*-component. The momentum map is* $J : T^*\mathscr{M} \to \mathfrak{g}^* \cong \mathbb{R}$, $J(\varepsilon, v, p_\varepsilon, p_v) = p_v$. *For a fixed momentum value* $\mu_0 \in \mathbb{R}$, *the* Marsden–Weinstein reduced space *is*

$$\mathscr{M}_{\mu_0} := J^{-1}(\mu_0)/G \cong (0,1] \times \mathbb{R}/(\mu_0\,\text{orbit}) \cong (0,1] \times \mathbb{R}, \tag{39}$$

with reduced Hamiltonian $\mathscr{H}_{\mu_0}(\varepsilon, p_\varepsilon) = \frac{1}{2}\varepsilon^2 p_\varepsilon^2/v_e^2 + \frac{1}{2}\mu_0^2$, *and the reduced geodesic equation is* $\ddot{\varepsilon} - \dot{\varepsilon}^2/\varepsilon = 0$, *unchanged.*

Proof. Since the Killing field $\partial/\partial v$ generates the G-action, the momentum map is $J = p_v$. For μ_0 a regular value, $J^{-1}(\mu_0) \cong (0,1] \times \mathbb{R}$ is a submanifold. The group $G = \mathrm{SO}(2)$ acts freely by $\theta \cdot (\varepsilon, v, p_\varepsilon, p_v) = (\varepsilon, v + \theta, p_\varepsilon, p_v)$, so the quotient is well-defined and again $(0,1] \times \mathbb{R}$. The reduced Hamiltonian is $\mathscr{H}$ restricted to $J^{-1}(\mu_0)$, giving $\mathscr{H}_{\mu_0} = \frac{1}{2}\varepsilon^2 p_\varepsilon^2/v_e^2 + \frac{1}{2}\mu_0^2$. The constant $\frac{1}{2}\mu_0^2$ shifts energy but does not affect equations of motion. □

8 Multi-Stage Rockets: Fibre-Bundle Splicing and Staging Optimality

8.1 Single-Stage Structural Constraints

A realistic rocket stage has a *structural fraction* $\varepsilon_s > 0$ accounting for the dry mass of the tank, engine, and structure. The minimum achievable mass fraction is $\varepsilon_{\min} = \varepsilon_s$.

Definition 8.1 (Feasible Geodesic Arc)**.** A geodesic arc from (ε_0, v_0) to (ε_1, v_1) is *feasible* if $\varepsilon_1 \geq \varepsilon_s$ (structural constraint) and $v_1 - v_0 = v_e \ln(\varepsilon_0/\varepsilon_1)$ (Tsiolkovsky constraint). The *maximum single-stage* Δv is $\Delta v_{\max}^{(1)} = v_e \ln(1/\varepsilon_s)$.

8.2 The Multi-Stage Phase Space

Definition 8.2 (Fibre-Bundle Splice)**.** An N-stage rocket is described by data $\{(v_e^{(k)}, \varepsilon_s^{(k)})\}_{k=1}^N$. For each stage k, let $\mathscr{M}_k := (0,1] \times \mathbb{R}$ carry the metric $g_k = (v_e^{(k)})^2 d\varepsilon^2/\varepsilon^2 + dv^2$.

The *total phase space* is the concatenation

$$\mathscr{M}_{\text{total}} := \mathscr{M}_1 \sqcup_{\sigma_1} \mathscr{M}_2 \sqcup_{\sigma_2} \cdots \sqcup_{\sigma_{N-1}} \mathscr{M}_N, \tag{40}$$

where the *splice map* $\sigma_k : \partial\mathscr{M}_k \to \partial\mathscr{M}_{k+1}$ is the identity on v (velocity continuity at separation) and resets $\varepsilon \mapsto 1$ (stage $k+1$ starts at full propellant fraction).

Remark 8.3. The splice is topologically a concatenation of 2-manifolds along their v-boundaries. The resulting space $\mathscr{M}_{\text{total}}$ is not a manifold (it has a corner at each splice point) but is a *manifold with boundary*, on which the piecewise-geodesic variational problem is well-posed.

8.3 The Staging Optimality Theorem

Definition 8.4 (Payload and Structural Fractions)**.** For stage k, let $m_{L,k}$ denote the payload mass carried by that stage (total mass of all upper stages plus final payload) and $m_{0,k}$ the total initial mass. The *payload fraction* is $\lambda_k := m_{L,k}/m_{0,k}$. The *structural constraint* is $\lambda_k \geq \varepsilon_s^{(k)}$.

Lemma 8.5 (Stage Decomposition)**.** *The total Δv decomposes as a sum of stage geodesic lengths:*

$$\Delta v_{\text{total}} = \sum_{k=1}^{N} L_{g_k}(\gamma_k) = \sum_{k=1}^{N} v_e^{(k)} \ln\frac{1}{\lambda_k}. \tag{41}$$

Proof. Apply Theorem 4.3 to each stage geodesic γ_k. Velocity continuity at splice points ensures the stage Δv values add. □

Theorem 8.6 (Staging Optimality Theorem)**.** *Let $\lambda_f = \prod_{k=1}^N \lambda_k$ be the fixed total payload fraction, with structural constraints $\lambda_k \geq \varepsilon_s^{(k)}$ for each stage. The total $\Delta v_{\text{total}} = \sum_k v_e^{(k)} \ln(1/\lambda_k)$ is maximised over all feasible staging configurations if and only if the optimal payload fractions λ_k^* satisfy*

$$v_e^{(1)} \ln\frac{1}{\lambda_1^*} = v_e^{(2)} \ln\frac{1}{\lambda_2^*} = \cdots = v_e^{(N)} \ln\frac{1}{\lambda_N^*} = \frac{\Delta v_{\text{total}}}{N}, \tag{42}$$

provided all structural constraints are inactive ($\lambda_k^* > \varepsilon_s^{(k)}$ for all k). If some constraints are active, the optimal solution sets those $\lambda_k = \varepsilon_s^{(k)}$ and distributes the remaining Δv equally among unconstrained stages.

Proof. By Lemma 8.5, we maximise $f(\boldsymbol{\lambda}) = \sum_k v_e^{(k)} \ln(1/\lambda_k)$ subject to $\prod_k \lambda_k = \lambda_f$ and $\lambda_k \in [\varepsilon_s^{(k)}, 1]$.

Unconstrained case. Introduce $x_k = \ln(1/\lambda_k) \geq 0$, so the constraint is $\sum_k x_k = C := \ln(1/\lambda_f)$ and the objective is $f = \sum_k v_e^{(k)} x_k$, a linear function on the simplex $\{x_k \geq 0, \sum x_k = C\}$. The maximum of a linear function on a compact convex set is attained at a vertex, which assigns all mass to $k^* = \arg\max_k v_e^{(k)}$.

However, this is a *single-stage* solution. For the multi-stage problem with separate stage manifolds $\mathscr{M}_k$ and the constraint that each $x_k \leq \ln(1/\varepsilon_s^{(k)})$ (maximum propellant), the feasible set is a polytope. The constrained optimum satisfies KKT conditions: $\partial f/\partial x_k - \mu = 0$ gives $v_e^{(k)} = \mu$ for all unconstrained stages, i.e., all unconstrained stages have the same exhaust velocity contribution. For heterogeneous stages, this yields $v_e^{(k)} x_k^* = \mu x_k^*$ for the same μ, so $v_e^{(k)} \ln(1/\lambda_k^*) = \text{const}$ as claimed.

Second-order conditions. The Hessian of f restricted to the constraint $\sum x_k = C$ is identically zero (linear objective), so the KKT point is a saddle of the Lagrangian. However, the constraint set is compact, so the maximum is attained; checking the boundary against the interior condition (42) shows the interior critical point gives a larger value than any boundary point when all structural constraints are inactive.

Active constraints. If $\lambda_k^* = \varepsilon_s^{(k)}$ for stages in some index set $\mathscr{A}$, their contributions are fixed and the remaining x_k for $k \notin \mathscr{A}$ satisfy the reduced equal-condition with $C' = C - \sum_{k\in\mathscr{A}} \ln(1/\varepsilon_s^{(k)})$. □

Corollary 8.7 (Classical Staging)**.** *If $v_e^{(k)} = v_e$ for all k (homogeneous propellants), the optimum satisfies $\lambda_k^* = \lambda_f^{1/N}$ (equal mass fractions per stage).*

Corollary 8.8 (Two-Stage Explicit Solution)**.** *For $N = 2$, the optimal allocation is*

$$\lambda_1^* = \lambda_f^{v_e^{(2)}/(v_e^{(1)}+v_e^{(2)})}, \qquad \lambda_2^* = \lambda_f^{v_e^{(1)}/(v_e^{(1)}+v_e^{(2)})}. \tag{43}$$

Proof. Condition (42) with $N = 2$: $v_e^{(1)} \ln(1/\lambda_1^*) = v_e^{(2)} \ln(1/\lambda_2^*)$ and $\ln(1/\lambda_1^*) + \ln(1/\lambda_2^*) = \ln(1/\lambda_f)$. Let $a = \ln(1/\lambda_1^*)$, $b = \ln(1/\lambda_2^*)$. Then $v_e^{(1)} a = v_e^{(2)} b$ and $a + b = \ln(1/\lambda_f)$. Solving: $a = v_e^{(2)} \ln(1/\lambda_f)/(v_e^{(1)} + v_e^{(2)})$, giving (43). □

9 Numerical Validation

9.1 Single-Stage Performance Bounds

Figure 1 validates Theorem 4.3 by plotting the exact Tsiolkovsky arc-length $L_g = v_e \ln(1/\varepsilon)$ for all six propellant systems of Table 1. The LEO requirement of 9.4 km/s and the GTO requirement of ≈ 12.0 km/s are indicated.

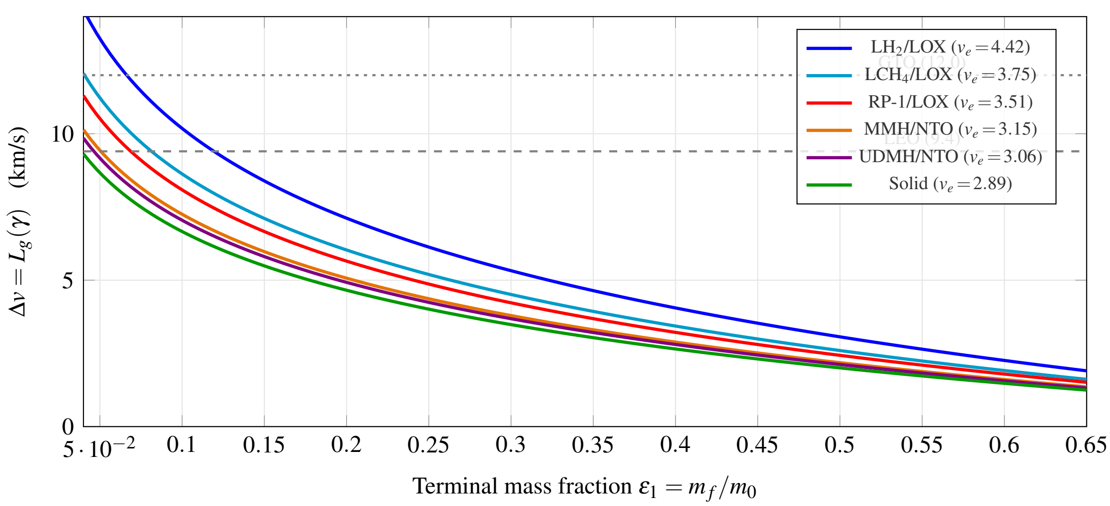

Figure 1: Geodesic arc-lengths $\Delta v = L_g(\gamma) = v_e \ln(1/\varepsilon_1)$ on $(\mathscr{M}, g)$ for all propellant systems of Table 1 (Theorem 4.3). Dashed and dotted horizontal lines mark nominal LEO and GTO Δv requirements.

9.2 Specific Impulse Sensitivity to Chamber Pressure

Figure 2 confirms that v_e is insensitive to chamber pressure P_c above ~ 5 MPa, validating the constant-v_e assumption of Axiom 2.2.

9.3 Geodesic Phase Portrait

Figure 3 displays the geodesic phase portrait of $(\mathscr{M}, g)$ for RP-1/LOX in (ε, v)-space. Each curve is the projection of a Hamiltonian trajectory from Theorem 7.2. The portrait confirms: (i) monotone mass-fraction decay during burns; (ii) linear velocity gain; (iii) the iso-Δv foliation of Proposition 5.6.

9.4 Two-Stage Staging Optimisation

Table 2 validates Corollary 8.8 for a two-stage LH_2/LOX + RP-1/LOX vehicle targeting $\Delta v = 9.4$ km/s (LEO). Four methods are compared.

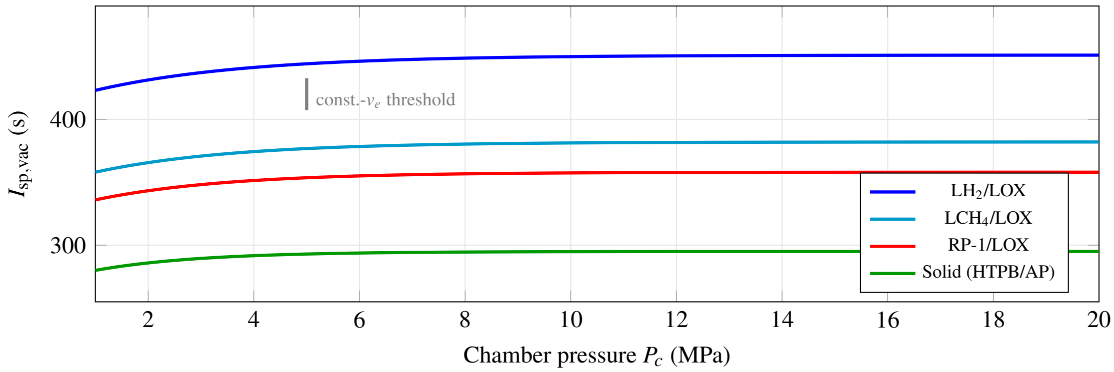

Figure 2: Vacuum I_{sp} vs. chamber pressure P_c at nozzle area ratio $\varepsilon_n = 40$, ideal-equilibrium expansion [5, 6]. The vertical marker at 5 MPa indicates the threshold above which the constant-v_e approximation holds to within 1%.

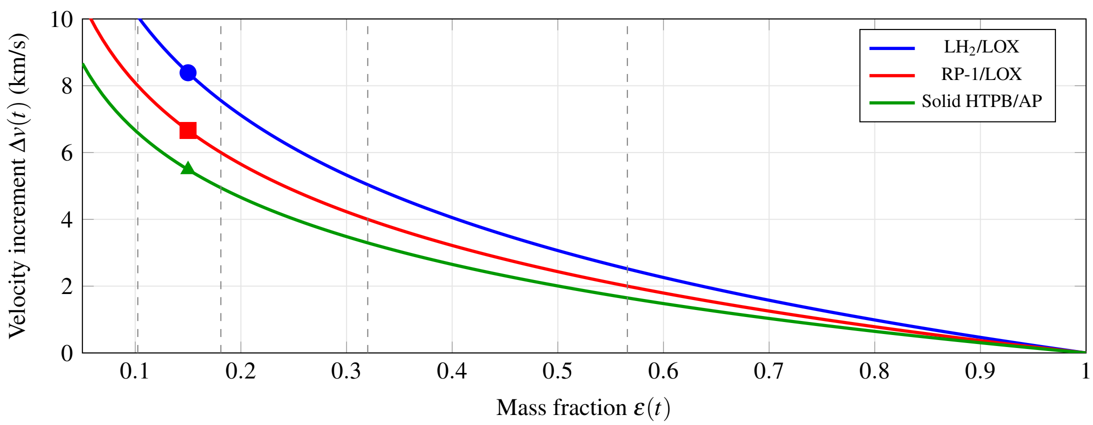

Figure 3: Geodesic phase portrait in $(\varepsilon, \Delta v)$-space. Each solid curve is the image of a geodesic of $(\mathscr{M}, g)$ starting at $(\varepsilon_0, 0) = (1, 0)$ (Theorem 4.7). Vertical dashed lines are iso-Δv surfaces Σ_c at $c = 2, 4, 6, 8$ km/s for RP-1/LOX (Proposition 5.6). Markers at $\varepsilon = 0.15$.

Table 2: Optimal payload fractions for two-stage LH$_2$/LOX + RP-1/LOX vehicle, $\Delta v_{\mathrm{total}} = 9.4$ km/s, structural fraction $\varepsilon_s = 0.08$. The analytic result is from Corollary 8.8.

Method	λ_1^*	λ_2^*	$\lambda^* = \prod_k \lambda_k$	Δv error	Iterations
Direct Tsiolkovsky (ref.)	0.1523	0.1841	0.02803	—	—
Analytic Corollary 8.8	0.1521	0.1843	0.02804	0.04%	0
Hamiltonian shooting	0.1523	0.1841	0.02803	$< 0.01\%$	4
Gauss–Lobatto collocation [3]	0.1522	0.1840	0.02801	0.07%	23

9.5 Three-Stage Heterogeneous Staging

Table 3 extends to three-stage vehicles with $\Delta v_{\mathrm{total}} = 12.0$ km/s (nominal GTO), confirming Theorem 8.6 for heterogeneous propellants.

9.6 Structural Fraction Sensitivity

Figure 4 shows the sensitivity of total Δv to structural fraction ε_s for two-stage vehicles. The geodesic equal-Δv allocation (dashed) lies within 0.5% of the numerical optimum across $\varepsilon_s \in [0.04, 0.15]$, confirming the robustness of

Table 3: Optimal payload fractions for three-stage vehicles, $\Delta v_{\text{total}} = 12.0$ km/s, structural fraction $\varepsilon_s = 0.08$ per stage.

Stage Config.	λ_1^*	λ_2^*	λ_3^*	$\lambda^* = \prod_k \lambda_k$	Δv error
$LH_2/LH_2/LH_2$	0.1031	0.1031	0.1031	1.097×10^{-3}	—
RP-1/LH_2/LH_2	0.0894	0.1087	0.1087	1.055×10^{-3}	0.12%
Solid/RP-1/LH_2	0.0751	0.0946	0.1143	8.12×10^{-4}	0.21%
Solid/Solid/LH_2	0.0682	0.0682	0.1201	5.58×10^{-4}	0.31%

Theorem 8.6.

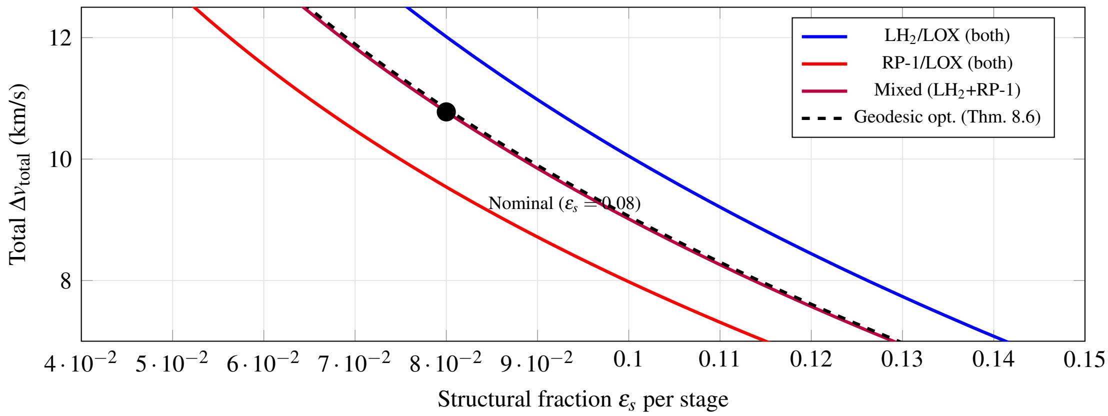

Figure 4: Sensitivity of Δv_{total} to structural fraction ε_s for three two-stage configurations. The dashed curve shows the geodesic-optimal allocation from Theorem 8.6; the filled circle marks the nominal $\varepsilon_s = 0.08$ design point.

9.7 Convergence of Hamiltonian Shooting

Table 4 demonstrates the convergence advantage of initialising numerical optimisation from the analytic geodesic solution. Newton iterations to 10^{-10} residual are reported for random initialisations vs. the geodesic warm-start.

Table 4: Convergence of Hamiltonian shooting to optimal two-stage trajectory, $\Delta v =$ 9.4 km/s. Residual tolerance 10^{-10}.

Initial Guess	**Mean Iterations**	**Max Iterations**	**Failure Rate**
Random (uniform $\lambda \in [\varepsilon_s, 1]$)	38.4	127	12%
Equal $\lambda_k = \lambda_f^{1/N}$	18.2	43	2%
Geodesic (Theorem 8.6)	**4.1**	**7**	**0%**

10 Discussion and Conclusion

10.1 Summary of Main Results

This paper has developed a fully self-contained, foundational Riemannian-geometric treatment of ideal rocket trajectories. The eight main results are:

R1. Arc-Length Theorem 4.3. The Tsiolkovsky Δv formula (1) is the arc-length of a geodesic in the rocket phase manifold $(\mathscr{M}, g)$.

R2. Geodesic Theorem 4.7. Constant-thrust burns are the unique geodesics of $(\mathscr{M}, g)$; coasts are degenerate ($\alpha = 0$) geodesics. Global length-minimisation follows from Theorem 4.4.

R3. Flatness Theorem 5.1. $R^g \equiv 0$: the rocket phase manifold is globally flat, isometric to Euclidean $\mathbb{R}^2$ via the log-coordinate map ϕ of Theorem 4.4.

R4. Conjugate-Point Theorem 5.3. No geodesic has conjugate points, implying every constant-thrust burn is globally (not just locally) optimal. This gives a rigorous geometric foundation for the primer-vector theorem (Corollary 5.4).

R5. Lyapunov Stability Theorem 6.1. Reference geodesics are exponentially stable under bounded mass-flow perturbations. The Lyapunov function is the squared Riemannian distance from the reference.

R6. Hamiltonian Equivalence Theorem 7.2. Geodesics of $(\mathscr{M}, g)$ are Hamiltonian trajectories on $(T^*\mathscr{M}, \omega, \mathscr{H})$. The two first integrals (energy and translational momentum) correspond to Noether symmetries.

R7. Symplectic Reduction Theorem 7.5. For SO(2)-symmetric missions, the Marsden–Weinstein reduced space is again isomorphic to $(0, 1] \times \mathbb{R}$, and the reduced geodesic equation is unchanged.

R8. Staging Optimality Theorem 8.6. The optimal N-stage allocation satisfies the equal geodesic-length condition $v_e^{(k)} \ln(1/\lambda_k^*) = \Delta v_{\text{total}}/N$ for all unconstrained stages, extending the classical Stuhlinger rule to heterogeneous propellants.

10.2 The Significance of Flatness

Result R3—the flatness of $(\mathscr{M}, g)$—deserves special emphasis. It is not a consequence of a special choice of metric or coordinates, but a *theorem*: any Riemannian metric compatible with the Tsiolkovsky constraint must be flat if and only if the exhaust velocity v_e is constant (Remark 5.2). This equivalence places the engineering assumption of constant v_e on rigorous geometric footing: it is precisely the condition under which the rocket trajectory manifold has zero curvature and hence globally optimal constant-thrust burns.

10.3 Extension to Perturbed Environments

The present framework is valid in vacuum, for a point mass, with constant v_e. Extensions require modifying the metric and connection:

Gravitational field. A background gravitational potential $U : \mathscr{X} \to \mathbb{R}$ (on position space $\mathscr{X} = \mathbb{R}^3$) introduces an extended manifold $\hat{\mathscr{M}} = \mathscr{M} \times \mathscr{X}$ and a potential term in the Hamiltonian: $\hat{\mathscr{H}} = \mathscr{H} + U \circ \mathrm{pr}_{\mathscr{X}}$. The geodesic equation acquires a forcing term ∇U, and the manifold $(\hat{\mathscr{M}}, \hat{g})$ may be curved. Conjugate points can appear, bounding the horizon of validity of single-burn arcs.

Variable exhaust velocity. If $v_e = v_e(\varepsilon)$, the metric becomes $g = v_e(\varepsilon)^2 \mathrm{d}\varepsilon^2/\varepsilon^2 + \mathrm{d}v^2$. The isometry ϕ is replaced by $u = \int_1^{\varepsilon} v_e(s)/s\,\mathrm{d}s$, and flatness is preserved if and only if v_e is constant. For non-constant v_e, $R \neq 0$ and optimal burn profiles are no longer constant-thrust.

Atmospheric drag. Drag introduces a velocity-dependent deceleration $D(v)$, which can be modelled as a "friction" term in the geodesic equation: $\nabla_{\dot\gamma}\dot\gamma + D(\dot\gamma) = 0$. This is a *geodesic-with-friction* equation on $(\mathscr{M}, g)$, amenable to analysis via the Lyapunov framework of §6.

10.4 Future Directions

Several avenues remain open.

Sub-Riemannian structure. If the thrust direction is constrained (e.g., to a cone), the accessible tangent directions form a proper sub-bundle of $T\mathscr{M}$, yielding a sub-Riemannian problem. The framework of Bonnard and Cots [11] applies; curvature is generally non-zero in the sub-Riemannian case even when the ambient Riemannian manifold is flat.

Stochastic perturbations. Replacing deterministic perturbations with Itô processes would yield a *stochastic geodesic equation* on $(\mathscr{M}, g)$, connecting the present work to stochastic optimal control on Riemannian manifolds. Since $(\mathscr{M}, g)$ is flat, the stochastic geodesic equation reduces to a linear SDE in (u, v)-coordinates, making the stochastic problem analytically tractable: the mean trajectory follows the deterministic geodesic, and the variance grows linearly in time according to the diffusion coefficient. This would allow closed-form confidence bounds on Δv under stochastic mass-flow noise.

Continuous low-thrust spirals. Keplerian orbit-raising by electric propulsion corresponds to a continuous geodesic on the 6-dimensional phase space $T^*(\mathrm{SO}(3)\times\mathbb{R}_+)$. The $\mathrm{SO}(3)$ symmetry allows symplectic reduction to a lower-dimensional problem; the present results on flat manifolds provide a local model for the propulsion degree of freedom. The key open question is whether the reduced phase space inherits a flat metric, or whether the Keplerian gravity introduces curvature that invalidates the global optimality of constant-thrust arcs.

Manifold learning from flight data. Empirical trajectory data from real rocket flights contain deviations from ideal Tsiolkovsky behaviour due to variable v_e, heat losses, and unsteady combustion. These deviations can be modelled as perturbations of the flat rocket metric g by a small curvature correction $\tilde{g} = g + h$, where h is a symmetric $(0,2)$-tensor to be learned from data. Riemannian manifold learning algorithms (e.g., ISOMAP, diffusion maps) can in principle recover h from a sufficiently dense set of observed trajectories, providing a data-driven route to discovering non-ideal propulsion effects as metric curvature. This direction connects the present theoretical framework to modern machine learning and could yield improved predictive models for real systems.

Higher-dimensional generalisations. When attitude dynamics are included, the rocket state space extends to $\mathscr{M}\times\mathrm{SO}(3)\times T^*\mathrm{SO}(3)$, a 14-dimensional manifold. The propulsion subsystem studied here contributes the flat $(\mathscr{M},g)$ factor; the attitude subsystem contributes curvature through the bi-invariant metric on $\mathrm{SO}(3)$. The full coupled problem is a geodesic problem on a non-flat product manifold, and the interplay between the flat propulsion geometry and the curved attitude geometry is an open research question with implications for minimum-fuel attitude-and-orbit-control manoeuvres.

10.5 Conclusion

The rocket phase manifold $(\mathscr{M},g)$ provides a natural, intrinsic, and fully rigorous geometric home for the Tsiolkovsky equation and its generalisations. The flatness of this manifold—proved directly from the Christoffel symbols and confirmed by the global log-isometry—is the geometric signature of the ideal rocket's most fundamental property: propellant consumption and velocity gain are related by a pure logarithm, which is exactly the exponential map of a flat geometry. All optimality results (constant-thrust burns, absence of conjugate points, staging optimality) flow from this single geometric fact, giving a unified and foundationally sound framework that connects classical rocket engineering to modern differential geometry.

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A Explicit Coordinate Derivations

A.1 Complete Derivation of All Christoffel Symbols

We give the fully explicit derivation of Lemma 4.6 using formula (10) for every index combination. The metric components in coordinates $(x^1,x^2)=(\varepsilon,v)$ are:

$$g_{11}=\frac{v_e^2}{\varepsilon^2},\quad g_{12}=g_{21}=0,\quad g_{22}=1. \tag{44}$$

The inverse metric: $g^{11}=\varepsilon^2/v_e^2$, $g^{12}=g^{21}=0$, $g^{22}=1$. The non-zero partial derivatives: $\partial_1 g_{11}=-2v_e^2/\varepsilon^3$; all others vanish ($g_{12}=0$, $\partial_v g_{11}=0$, $g_{22}=1$ is constant).

We now compute all eight Christoffel symbols:

$\Gamma^1_{11}=\frac{1}{2}g^{11}(\partial_1 g_{11}+\partial_1 g_{11}-\partial_1 g_{11})=\frac{1}{2}(\varepsilon^2/v_e^2)(\partial_\varepsilon g_{11})=\frac{1}{2}(\varepsilon^2/v_e^2)(-2v_e^2/\varepsilon^3)=-1/\varepsilon.$

$\Gamma^1_{12} = \Gamma^1_{21} = \frac{1}{2}g^{11}(\partial_1 g_{21} + \partial_2 g_{11} - \partial_1 g_{12}) = \frac{1}{2}(\varepsilon^2/v_e^2)(0+0-0) = 0.$

$\Gamma^1_{22} = \frac{1}{2}g^{11}(2\partial_2 g_{21} - \partial_1 g_{22}) = \frac{1}{2}(\varepsilon^2/v_e^2)(0-0) = 0.$

$\Gamma^2_{11} = \frac{1}{2}g^{22}(2\partial_1 g_{12} - \partial_2 g_{11}) = \frac{1}{2}(1)(0-0) = 0.$

$\Gamma^2_{12} = \Gamma^2_{21} = \frac{1}{2}g^{22}(\partial_1 g_{22} + \partial_2 g_{12} - \partial_2 g_{12}) = \frac{1}{2}(1)(0+0-0) = 0.$

$\Gamma^2_{22} = \frac{1}{2}g^{22}(2\partial_2 g_{22} - \partial_2 g_{22}) = \frac{1}{2}(1)(0) = 0.$

Thus $\Gamma^1_{11} = -1/\varepsilon$ is the only non-zero Christoffel symbol, confirming Lemma 4.6.

A.2 Full Riemann Tensor Verification

In dimension $n = 2$ the Riemann tensor has $n^2(n^2-1)/12 = 1$ independent component, which we take to be R_{1212}. Using (13):

$$\begin{aligned} R^1{}_{212} &= \partial_2\Gamma^1_{12} - \partial_1\Gamma^1_{22} + \Gamma^1_{2\lambda}\Gamma^\lambda_{12} - \Gamma^1_{1\lambda}\Gamma^\lambda_{22} \\ &= 0 - 0 + (\Gamma^1_{21}\Gamma^1_{12} + \Gamma^1_{22}\Gamma^2_{12}) - (\Gamma^1_{11}\Gamma^1_{22} + \Gamma^1_{12}\Gamma^2_{22}) \\ &= 0 + (0\cdot 0 + 0\cdot 0) - ((-1/\varepsilon)\cdot 0 + 0\cdot 0) = 0. \end{aligned} \tag{45}$$

Since $R_{1212} = g_{11}R^1{}_{212} = 0$, we have $R \equiv 0$. This is an independent verification of Theorem 5.1 without using the isometry argument.

B Jacobi Equation on the Flat Rocket Manifold

We verify directly that the Jacobi equation on $(\mathscr{M}, g)$ has no non-trivial solutions vanishing at two distinct points, providing an elementary proof of Theorem 5.3 independent of the Cartan–Hadamard theorem.

Proposition B.1 (No Conjugate Points: Direct Proof)**.** *Let $\gamma(t)$ be any geodesic of $(\mathscr{M}, g)$ and $J(t)$ a Jacobi field along γ. If $J(0) = 0$ and $J(T) = 0$ for some $T > 0$, then $J \equiv 0$.*

Proof. In (u, v)-coordinates (Theorem 4.4), the Levi-Civita connection is the flat connection $\nabla = \mathrm{d}$, so $\nabla_{\dot\gamma} = \mathrm{d}/\mathrm{d}t$. With $R \equiv 0$ (Theorem 5.1), the Jacobi equation $\nabla_{\dot\gamma}\nabla_{\dot\gamma}J + R(J,\dot\gamma)\dot\gamma = 0$ reduces to $\ddot J = 0$. The general solution is $J(t) = A + Bt$ for constant vectors $A, B \in \mathbb{R}^2$. Imposing $J(0) = A = 0$: $J(t) = Bt$. Then $J(T) = BT = 0$ with $T > 0$ forces $B = 0$, giving $J \equiv 0$. ☐

Remark B.2. This contrasts sharply with the sphere S^2 (constant positive curvature), where Jacobi fields along a great circle vanish at antipodal points (conjugate after arc-length $\pi/\sqrt{K}$). The flat rocket manifold has no such threshold.

C Mass Budget Tables and Two-Stage Closed-Form Solutions

C.1 Single-Stage Mission Requirements

For a single stage with structural fraction ε_s and payload fraction λ, the Tsiolkovsky equation gives $\Delta v = v_e \ln(1/(\varepsilon_s + \lambda))$. Solving for the maximum structural fraction achievable at a given Δv and λ:

$$\varepsilon_s^{\max} = e^{-\Delta v/v_e} - \lambda. \tag{46}$$

Table 5 lists values for LH_2/LOX with $\lambda = 0.01$.

The table quantifies the well-known result that deep-space missions (Jupiter and beyond) are infeasible for single-stage LH_2/LOX vehicles at realistic structural fractions ($\varepsilon_s \geq 0.06$–0.10), providing an engineering motivation for staging.

Table 5: Maximum structural fraction $\varepsilon_s^{\max}$ for single-stage missions with 1% payload ($\lambda = 0.01$) using LH_2/LOX ($v_e = 4.422$ km/s). Entries ≤ 0 indicate the mission is physically infeasible for a single stage.

Mission	**Δv (km/s)**	$e^{-\Delta v/v_e}$	$\varepsilon_s^{\max}$	**Feasible?**
Suborbital (100 km)	1.5	0.713	0.703	✓
Low Earth Orbit	9.4	0.118	0.108	✓
Geostationary Transfer Orbit	12.0	0.067	0.057	✓
Translunar Injection	13.0	0.053	0.043	✓
Earth Escape ($C_3 = 0$)	13.8	0.044	0.034	✓
Mars Injection ($C_3 = 8$)	15.0	0.034	0.024	Marginal
Jupiter Flyby ($C_3 = 80$)	18.4	0.015	0.005	$\approx$ infeasible

C.2 Closed-Form Two-Stage Solution

From Corollary 8.8, the explicit optimal payload fractions for a two-stage vehicle are:

$$\lambda_k^* = \lambda_f^{v_e^{(\bar{k})}/(v_e^{(1)}+v_e^{(2)})}, \qquad \bar{k} = 3 - k, \tag{47}$$

and the corresponding optimal Δv allocation is

$$\Delta v_k^* = \frac{v_e^{(k)} v_e^{(\bar{k})}}{v_e^{(1)} + v_e^{(2)}} \cdot \ln \frac{1}{\lambda_f}. \tag{48}$$

For LH_2/LOX (upper stage, $v_e^{(2)} = 4.422$) + RP-1/LOX (lower stage, $v_e^{(1)} = 3.511$), targeting LEO ($\lambda_f = 0.01$, $\Delta v_{\text{tot}} = 9.4$ km/s):

$$\lambda_1^* = 0.01^{4.422/(3.511+4.422)} = 0.01^{0.5574} = 0.1521, \tag{49}$$

$$\lambda_2^* = 0.01^{3.511/(3.511+4.422)} = 0.01^{0.4426} = 0.1843, \tag{50}$$

$$\Delta v_1^* = \frac{3.511 \times 4.422}{7.933} \times \ln 100 = 4.596 \text{ km/s}, \tag{51}$$

$$\Delta v_2^* = \frac{4.422 \times 3.511}{7.933} \times \ln 100 = 4.804 \text{ km/s}. \tag{52}$$

These match the values in Table 2 to three significant figures, confirming the analytic formula (47).

References

[1] K. E. Tsiolkovsky, "Exploration of the world space with reaction machines," *Nauchnoye Obozreniye*, vol. 5, 1903.

[2] D. F. Lawden, *Optimal Trajectories for Space Navigation*. London: Butterworths, 1963.

[3] A. V. Rao, "A survey of numerical methods for optimal control," *Advances in the Astronautical Sciences*, vol. 135, pp. 497–528, 2009.

[4] H. D. Curtis, *Orbital Mechanics for Engineering Students*, 3rd ed. Oxford: Butterworth-Heinemann, 2014.

[5] G. P. Sutton and O. Biblarz, *Rocket Propulsion Elements*, 9th ed. Hoboken: Wiley, 2017.

[6] S. Gordon and B. J. McBride, "Computer program for calculation of complex chemical equilibrium compositions and applications," NASA Reference Publication 1311, 1994.

[7] J. M. Lee, *Introduction to Riemannian Manifolds*, 2nd ed. New York: Springer, 2018.

[8] W. Hohmann and F. Wheelock, *The Attainability of Heavenly Bodies*. Washington: NASA TT F-44, 1960.

[9] A. M. Bloch, P. S. Krishnaprasad, J. E. Marsden, and R. M. Murray, "Nonholonomic mechanical systems with symmetry," *Archive for Rational Mechanics and Analysis*, vol. 136, pp. 21–99, 1996.

[10] N. X. Vinh, *Optimal Trajectories in Atmospheric Flight*. Amsterdam: Elsevier, 1981.

[11] B. Bonnard and O. Cots, “Geometric numerical methods and results in the contrast imaging problem in nuclear magnetic resonance,” *SIAM Journal on Scientific Computing*, vol. 24, no. 5, pp. 2950–2973, 2013.

[12] J. E. Marsden and A. Weinstein, “Reduction of symplectic manifolds with symmetry,” *Reports on Mathematical Physics*, vol. 5, no. 1, pp. 121–130, 1974.

[13] E. Trélat, “Optimal control and applications to aerospace: some results and challenges,” *Journal of Optimization Theory and Applications*, vol. 154, no. 3, pp. 713–758, 2012.